

CREDIT RISK SCORING – LOAN DEFAULT/RISK PREDICTION

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PROBLEM STATEMENT

**HOW CAN WE
ACCURATELY
ASSESS AND
PREDICT THE
CREDIT RISK OF
LOAN APPLICANTS
USING RELIABLE
BORROWER DATA?**



DATA QUALITY

Borrower data may contain missing, inconsistent, duplicate, and noisy records.

RELIABILITY

Poor-quality data can affect the accuracy of credit risk analysis.

EFFICIENCY

Manual data inspection becomes time-consuming for large datasets.

PREDICTION

A data-driven approach is needed to predict loan default risk and classify borrowers.

MOTIVATION



FINANCIAL LOSS

Reduce potential losses caused by loan defaults.



DATA QUALITY

Improve the reliability and consistency of borrower data.



AUTOMATION

Reduce manual data inspection and processing.



RISK PREDICTION

Use machine learning to identify borrowers likely to default.



DECISION MAKING

Support more informed and efficient lending decisions.



CENTRALIZATION

Combine data quality analysis and credit risk prediction in one platform.

OBJECTIVES

Data Integration

Collect and combine loan-related data from CSV/Excel files, APIs, and web sources.

Data Preparation

Clean the data by handling missing values, outliers, inconsistencies, duplicates, and noise.

Risk Prediction

Develop and compare Logistic Regression, Random Forest, and XGBoost models to predict loan default risk.

Risk Classification

Classify borrowers into Low, Medium, and High Risk categories based on predicted default probability.

Dashboard

Provide an interactive platform to visualize data quality, borrower characteristics, risk distribution, and model performance.

EXPECTED OUTCOMES

Reliable Data — Generate clean and consistent borrower datasets with improved data quality.

Accurate Prediction — Predict loan default probability and identify important factors influencing credit risk.

Risk Assessment — Classify borrowers into Low, Medium, and High Risk categories using evaluated ML models.

Interactive Dashboard — Visualize data quality, risk distribution, borrower information, and model performance.

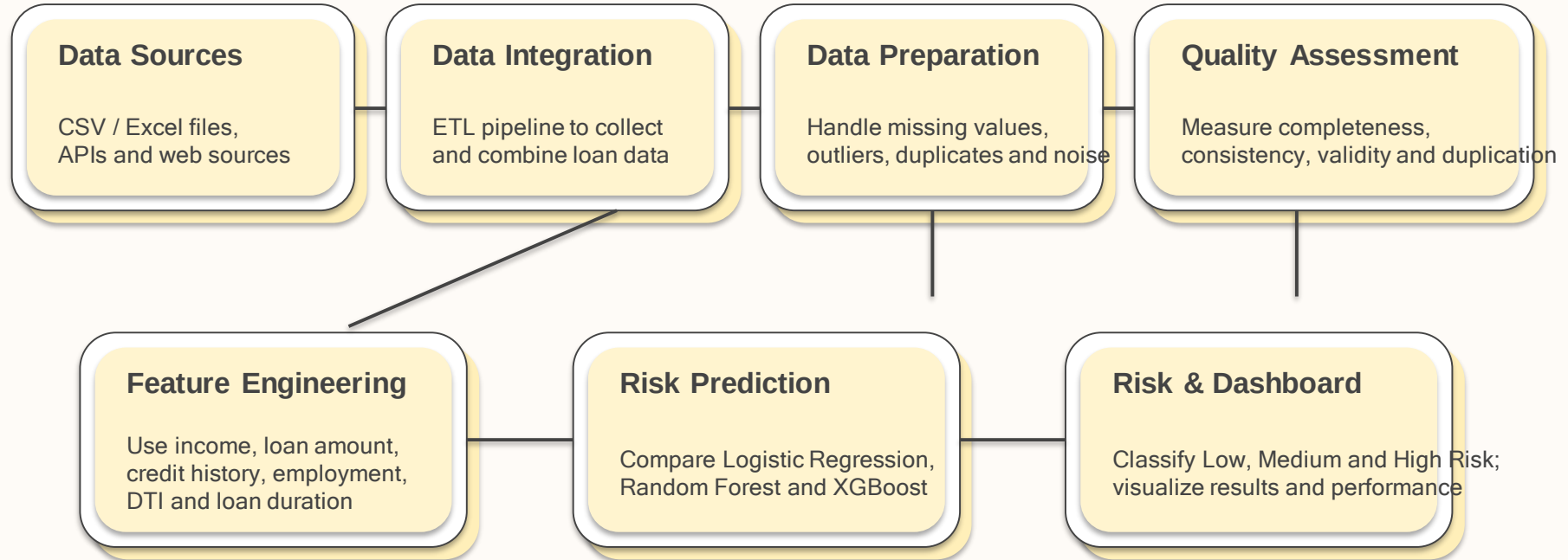


LITERATURE SURVEY / EXISTING SYSTEM

Title	Author	Year	Dataset Link	Limitation
Benchmarking Classification Algorithms for Credit Scoring	Stefan Lessmann, Bart Baesens, Hsin-Vonn Seow, Lyn C. Thomas	2015	https://www.sciencedirect.com/science/article/abs/pii/S0377221715004208?via%3Dihub	No focus on leakage or explainability
Predicting Good Probabilities with Supervised Learning	Alexandru Niculescu-Mizil, Rich Caruana	2005	https://explore.metascienceobservatory.org/papers/W2098824882	Not credit-risk specific
Financial Ratios, Discriminant Analysis and the Prediction of Corporate Bankruptcy	Edward I. Altman	1968	Corporate bankruptcy dataset — 66 firms used in the original study	Small, industry-specific dataset
LightGBM: A Highly Efficient Gradient Boosting Decision Tree	Guolin Ke et al.	2017	https://github.com/lightgbm-org/LightGBM/blob/main/docs/Experiments.rst	Not designed for credit risk
CatBoost: Unbiased Boosting with Categorical Features	Liudmila Prokhorenkova et al.	2018	https://catboost.ai/docs/en/concepts/python-reference_datasets	General-purpose, not credit-specific

PROPOSED SYSTEM

An integrated pipeline for reliable credit-risk analysis and loan default prediction.



Integrated data processing → machine learning → risk assessment → visualization

THANK YOU

